

Itô's Lemma FROM SCRATCH

Goal of these notes: Derive and understand Itô's Lemma

Reminder (from Ordinary Calculus):

• Total Derivative: $df(t, x) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx$

• Taylor Series: $f(x) = f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots =$
 $= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \rightarrow \int_a^b f'(t) dt \cdot f(b) \cdot f(a)$
+
integrating by parts

$\Rightarrow f(a + \Delta x) = f(a) + \frac{f'(a)}{1!} \Delta x + \frac{f''(a)}{2!} \Delta x^2 + \dots$

$\hookrightarrow f(\Delta x) = \frac{f'(a)}{1!} \Delta x + \frac{f''(a)}{2!} \Delta x^2 + \dots$ considering $\Delta x \rightarrow 0, \Delta x \rightarrow dx$

$\hookrightarrow \frac{\partial f}{\partial x} dx = \frac{1}{1!} \frac{\partial f(a)}{\partial x} dx + \frac{1}{2!} \frac{\partial^2 f(a)}{\partial x^2} (dx^2) + \dots$

$\Rightarrow$ Now we can express the total derivative of a function with respect to t and x using the Taylor expansion

$\hookrightarrow df(t, x) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx = \left(\frac{\partial f}{\partial t} dt + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} dt^2 + \dots \right) + \left(\frac{\partial f}{\partial x} dx + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} dx^2 + \dots \right)$
 $= \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx + \frac{1}{2} \left(\frac{\partial^2 f}{\partial t^2} dt^2 + \frac{\partial^2 f}{\partial x^2} dx^2 \right) + \dots$

$\Rightarrow$ Introducing Itô's calculus $x \rightarrow \bar{x}(t)$ (stochastic) $\leftarrow f(t, \bar{x})$

Itô: $d\bar{x}(t) = \underbrace{a(t, \bar{x}) dt}_{\text{Drift}} + \underbrace{b(t, \bar{x}) dW(t)}_{\text{Diffusion}}$

Brownian motion
 $dW \sim \mathcal{N}(\epsilon \sqrt{dt}, \Phi(\mu=0, \sigma=1))$
 $\hookrightarrow$ variance additive
std. not additive
 $\hookrightarrow$ THAT'S WHY UNDER SQUARE ROOT

$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial \bar{x}} (a dt + b dW) +$
 $+ \frac{1}{2} \left[\frac{\partial^2 f}{\partial t^2} dt^2 + \frac{\partial^2 f}{\partial \bar{x}^2} (a^2 dt^2 + 2ab dt dW + b^2 dW^2) \right]$

deriving the result when $dt \rightarrow 0$

$dW^2 = ? \rightarrow dW = \epsilon \sqrt{dt} \quad \Phi(\mu=0, \sigma=1) \rightarrow dW^2 = \epsilon^2 dt$

$\text{Var}(\epsilon) = \sigma^2 = 1 = \underbrace{E[\epsilon^2] - E[\epsilon]^2}_0 \Rightarrow E[\epsilon^2] = 1$

$E[dW^2] = E[\epsilon^2 dt] = dt E[\epsilon^2] = dt$

$\text{Var}[dW^2] = \text{Var}[\epsilon^2 dt] = \underbrace{E[(\epsilon^2 dt)^2]}_{dt^2 E[\epsilon^4]} - \underbrace{E[\epsilon^2 dt]^2}_{dt^2} = 2 dt^2$
 $\hookrightarrow dt \rightarrow 0$

negligible

HENCE

IT BECOMES A NON-STOCHASTIC PROCESS!

$\Rightarrow df = \left(\frac{\partial f}{\partial t} + a(t, \bar{x}) \frac{\partial f}{\partial \bar{x}} + \frac{1}{2} b^2(t, \bar{x}) \frac{\partial^2 f}{\partial \bar{x}^2} \right) dt + b(t, \bar{x}) \frac{\partial f}{\partial \bar{x}} dW(t)$

Itô's Lemma